



Budgeting & Household Expenses

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 Emma is the driver at fault in a car accident. Which of the following is covered by Emma's compulsory third-party (CTP) insurance? 1
- A. Repairs to Emma's car
- B. Injury to the other driver**
- C. Damage to the other driver's car
- D. Cost of repairing a building damaged in the accident

- 2 The cost of a calculator, including 10% GST, is \$24.20. Calculate the GST. **\$2.20** 1

- 3 Andrew purchased a car for \$55 950. The table shows the stamp duty rates for a motor vehicle in the state where Andrew lives and purchased the car. Using the table, calculate the stamp duty payable on the car. 2

Vehicle value	Rate
0–\$4999	Nil
\$5000–\$44 999	\$3 for every \$100 (or part thereof)
\$45 000 and over	\$1300 plus \$6 for every \$100 (or part thereof)

To calculate stamp duty, the purchase price must be rounded up to the nearest \$100.

Purchase price 55 950 = \$56 000

$$1300 + 6 \times 56\,000 \div 100 = 4660$$

Therefore, stamp duty is \$4660

- 4 Sebastian has created a spreadsheet to help him budget for his fixed expenses each month.

- a. Calculate the monthly cost of electricity (X).

$$X = 2230 - (1560 + 250 + 85 + 185) = \$150$$

- b. The average cost of electricity throughout the year is 32c/kWh. Using this average, determine how many kWh Sebastian uses per year.

	A	B
1	Sebastian's Fixed Monthly Expenses	
2		
3	Electricity	X
4	Rent	\$1,560
5	Entertainment	\$250
6	Water	\$85
7	Fuel	\$185
8		
9	Total	\$2,230

$$\text{Monthly cost} = 150 \div 0.32 = 468.75 \text{ kWh}$$

$$\text{Yearly usage} = 468.75 \times 12 = 5625 \text{ kWh}$$

- 5 The rates for car stamp duty in NSW are shown in the table below. Calculate the stamp duty on a car valued at \$58 700. 2

Vehicle value	Rate
\$0 – \$44 999	\$3 for every \$100 (or part of \$100)
\$45 000 and over	\$1350 plus \$5 for every \$100 (or part of \$100) over \$45 000

$$1350 + 5 \left(\frac{58700 - 45000}{100} \right) = \$ 2035$$

- 6 Jacinta has a credit card that charges a simple interest rate of 21% per annum. The

	Transaction	Amount (\$)
13 January	Public transport card top-up	30.00
18 January	GK Drive driving lesson	89.00

credit card does not have an interest-free period, and interest is charged up to and including the date of payment. During one weekend, Jacinta makes the following transactions.

- a. Calculate the daily interest of Jacinta's credit card as a percentage. Give your answer correct to three decimal places. 1

$$\frac{21}{365} = 0.0575 \dots \approx 0.058\%$$

- b. Jacinta is due to pay her account in full on 9 February. What will be the total interest payable on Jacinta's transactions? 2

$$\begin{aligned} 13 \text{ Jan} - 9 \text{ Feb} &= (31 - 13 + 1) + 9 = 28 \text{ days} \\ 18 \text{ Jan} - 9 \text{ Feb} &= (31 - 18 + 1) + 9 = 23 \text{ days} \\ \text{total interest} &= (30 \times 28 \times 0.058\%) + (89 \times 23 \times 0.058\%) \\ &= 0.4872 + 1.18726 = 1.67446 \approx \$1.67 \end{aligned}$$

- 7 Remi purchased a car for \$34 850. 2

The table shows the stamp duty rates for a motor vehicle in the

Vehicle value	Rate
0–\$4999	Nil
\$5000–\$44 999	\$3 for every \$100 (or part thereof)
\$45 000 and over	\$1300 plus \$6 for every \$100 (or part thereof)

state where Remi lives and purchased the car. Using the table, calculate the stamp duty payable on the car.

To calculate stamp duty, the purchase price must be rounded up to the nearest \$100.

$$\text{purchase price} = 34850 \approx \$34900$$

$$\text{stamp duty} = 3\% \times 34900 = \$1047$$

- 8 A supermarket receipt is shown. Given that the cost of cat food tins is 3 times the cost of the ice-cream, determine the missing values P and Q to complete the receipt. 2

$$\begin{aligned} P + Q &= \$25.30 - \$3.50 \\ &= \$22 \end{aligned}$$

$$\begin{aligned} P + Q &= 3Q + Q \\ 22 &= 4Q \end{aligned}$$

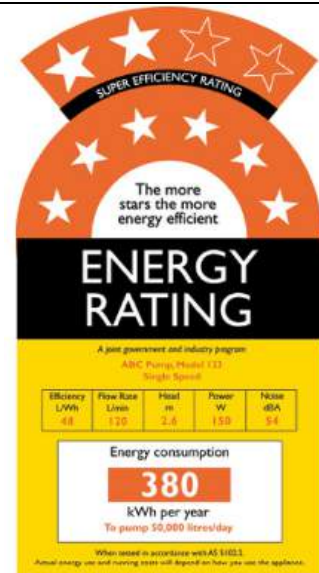
$$Q = \$5.50 \quad (\text{ice-cream})$$

$$3Q \text{ or } P = 3 \times 5.50$$

$$= \$16.50 \quad (\text{cat food tins})$$

BEST SUPERMARKET Receipt	
Bread	\$3.50
*Cat food tins	P
*Ice-cream	Q
Total for 3 items	\$25.50
GST included in total	\$3.30
*GST of 10% is included in the price of item.	

- 9 The cost of a new water pump is \$435. It uses energy according to the following energy label. Energy is charged at the rate of \$0.36 per kWh. How much will it cost in total to purchase and then run this pump for 6 months.



3

Cost of heater = \$435

Energy consumption = $380 \times 0.5 = 190$ kWh

Cost of energy = $190 \times 0.36 = \$68.40$

$\therefore$ Total cost to run heater = Purchase cost + Energy cost
 $= 435 + 68.40 = \$503.40$

- 10 Rebecca has a credit card with the following conditions:

- There is no interest free period.
- Interest at the rate of 0.04% per day is charged at the end of each month.
- Interest is calculated from and including the date of purchase to the last day of the month.

Statement period: 1 August to 31 August		
Date	Details	Amount (\$)
1 August	Opening balance	0
21 August	Fridge	4 500
31 August	Interest charge	
31 August	closing balance	
Minimum payment:		

3

Rebecca's credit card statement for August is shown, with some figures missing.

$$\begin{aligned}
 &\text{No. of days counted for interest} = 11 \\
 &I = 4500 \times 0.04\% \times 11 \\
 &= \$19.80 \\
 &\text{closing Balance} = \$4500 + \$19.80 \\
 &= \$4519.80 \\
 &\text{Minimum payment} = 0.05 \times \$4519.80 \\
 &= \$225.99
 \end{aligned}$$

- 11 Jas and Joe have a household fridge with an energy rating of 682 kWh per year. They are thinking of buying a bar fridge. The bar fridge they are considering cost \$560 and has an energy rating of 352 kWh per year. Their energy plan charges usage at 24.637c/kWh.

- a. What would be the total cost of purchasing and running the bar fridge for 3 years?

Cost of running bar fridge = $352 \times \$0.24637 = \86.72

Total cost for 3 years = $\$86.72 \times 3 + \$560 = \$820.17$

- b. What would be the percentage increase in cost of running both fridges compared to just the current fridge? (not including the purchase price)

Percentage increase = $\frac{352}{682} \times 100 = 51.6129... = 52\%$

- 12 A 400 W desktop computer is run on average for 4 hours per day. If electricity is charged at \$0.18/kWh, calculate the electricity cost of running the computer per year.

\$105.12

1

- 13 Which of the following is not covered by comprehensive car insurance in an accident? 1
- A. Injury to the other driver.
- B. Damage to the other driver's car.
- C. Repairs to own car.
- D. Cost of repairing a damaged building or property caused by an accident with own car.
- 14 A 3400-watt air conditioner is used to cool a room. The air conditioner is used for a duration of 105 minutes. If electricity costs 35.82 cents per kilowatt-hour, calculate the cost of electricity to cool the room in this time. Correct your answer to the nearest cent. 2
- 3400 watt = 3.4 kW
105 minutes = 1 hour and 45 minutes = 1.75 hours
35.82 cents = \$0.3582
Total Cost = $3.4 \times 1.75 \times \$0.3582 = \2.13
- 15 A 120 watt ceiling fan runs for 24 hours each day. If electricity is charged at 24.8c/kWh, what is the cost of running the ceiling fan for 30 days, to the nearest cent? 1
- \$21.43**



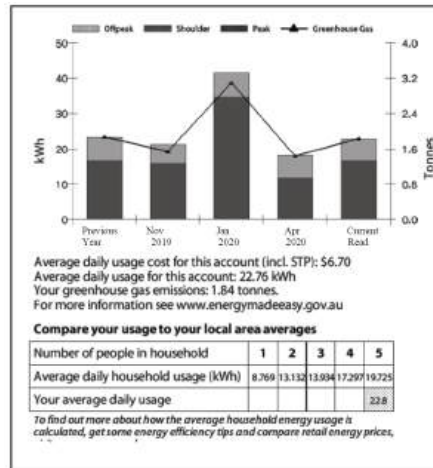
Budgeting & Household Expenses

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 A section of the electricity bill for Corie's share-house is shown below. There are five people living in the house who share the electricity costs



ISSUE DATE 16 AUG 20	
TRANSACTIONS SINCE PREVIOUS ACCOUNT (Incl GST)	
Previous Invoice Amount	\$500.04
Pay On Time™ Discount (incl GST of \$4.55)	\$50.00 Cr
Payment Received - Thank You	\$450.04 Cr
Balance Brought Forward	\$0.00
CURRENT TRANSACTIONS (Incl GST)	
Electricity Charges \$643.26	
TOTAL CURRENT TRANSACTIONS (Incl GST)	\$643.26
TOTAL AMOUNT DUE (Incl GST)	\$643.26
Total GST for this account is \$53.93	
10% PAY ON TIME™ DISCOUNT	\$64.33 Cr
If paid by 05 SEPT 20	
XXXXXXXX	
Total Amount Due with discount (Incl GST)	

2

equally. If she pays the account on 2nd September, how much would she need to collect from each of her housemates?

Paid in time to receive the discount,
 So amount due = $643.26 - 64.33 = \$578.93$
 Amount shared = $578.93 \div 5 = \$115.79$
 (Accept this answer or \$115.80)

- 2 Liza buys a car which has a market value of \$60 000 before on-road costs. Stamp duty on the car is calculated at these rates:

- 3% of the market value up to and including \$45,000
- 5% of the market value over \$45,000.

$$\begin{aligned} \text{Stamp duty} &= 3\% \text{ of } 45\,000 + 5\% \text{ of } 15\,000 \\ &= .03 \times 45\,000 + .05 \times 15\,000 \\ &= 1\,350 + 750 = \$2\,100 \\ \text{Cost of car} &= 60\,000 + 2\,100 = \$62\,100 \end{aligned}$$

How much will Liza pay for the car, including stamp duty.

- 3 Lily buys an LCD television which is rated at 80 watts when being used for viewing. When in stand-by mode it is rated at 4 watts. Lily and her family use the TV for viewing for 8 hours a day on average and leave it on standby for the remaining time.

- a. Show that the TV uses 0.704 kilowatt hours of energy on an average day.

$$\text{Usage} = \frac{80 \times 8 + 4 \times 16}{1000} = 0.704 \text{ kWh}$$

1

- b. Lily pays 30 cents per kWh for electricity. How much money would she save in a year if she turned the TV off when it wasn't being viewed, rather than leaving it on stand-by?

2

$$\begin{aligned} \text{Amount used in 1 day on standby for 16 hours} &= 4 \times 16 \div 1000 = 0.064 \text{ kWh} \\ \text{Amount used in a year} &= 0.064 \times 365 = 23.36 \text{ kWh} \\ \text{Money saved in a year} &= 23.36 \times 0.30 = \$7.008 = \$7.01 \text{ per year} \end{aligned}$$

- 4 The transactions for Donna's credit card are shown below for the month of August.

Credit Card Ms D M Snow			
	Opening Balance	\$0.00	
Date	Detail	Transaction	Balance
4-Aug-20	Woolworths DUBBO	-\$17.15	-\$17.15
16-Aug-20	SMITHS PHARMACY PARKES	-\$21.28	-\$38.43
16-Aug-20	Ampol Petroleum Parkes	-\$28.98	-\$67.41
20-Aug-20	COLES BATHURST	-\$14.90	-\$82.31
20-Aug-20	SUBWAY BATHURST	-\$22.65	-\$104.96
20-Aug-20	CHARLES STURT UNIVER BATHURST	-\$20.00	-\$124.96
20-Aug-20	Ampol Petroleum Orange	-\$34.53	-\$159.49
31-Aug-20	Officeworks ORANGE	-\$180.00	-\$339.49
31-Aug-20	NETFLIX.COM Melbourne	-\$13.99	-\$353.48
31-Aug-20	INTEREST DEBIT		

The bank charges

0.041% per day simple

interest on the

maximum daily balance on the card.

- a. Show that, in the first 15 days of the month, 8 cents in interest is charged. 1

No interest for first 3 days as zero balance

From 4th to 15th balance = 17.15

Interest = $17.15 \times 0.00041 \times 12 = \0.08

- b. How much interest is charged on the card for August? 2

From 16th to 19th balance = 67.41

Interest = $67.41 \times 0.00041 \times 4 = \0.11

From 20th to 30th balance = 159.49

Interest = $159.49 \times 0.00041 \times 11 = \0.72

On 31st balance = 353.48

Interest = $353.48 \times 0.00041 \times 1 = \0.14

Interest charged = $0.08 + 0.11 + 0.72 + 0.14 = \1.05

- 5 Adi buys a new heater which is rated at 2 000 watts. He uses it on 120 days of the year for an average of 8 hours per day. Electricity costs are \$0.30 per kilowatt hour. What would the heater cost to run for the year? 2

The heater is 2 000 watts which is 2 kw.

It is run for $120 \times 8 = 960$ hours per year

Power used = $960 \times 2 = 1920$ kwh

Cost at \$0.30 per kwh = $1920 \times 0.3 = \$576$

- 6 Torey lists the costs of running his car: 1

- Registration \$378 annually
- CTP insurance \$546.00 annually
- Comprehensive Insurance \$460.00 annually
- Mechanical Service/Repairs \$450.00 every three months
- Petrol and Consumables \$40.00 weekly

He wants to put money into an account each week to cover these costs and allow an extra 10% contingency in case of unforeseen expenses. How much (to the nearest dollar) should he deposit each week?

\$111.00

- 7 Amelia uses an 1800 W dishwasher for a total of 21 hours during the week. What is the cost of using the dishwasher for a week, if electricity is \$0.24 per kWh? 2

$$\text{Energy used} = 1800 \times 21 = 37800 \text{ Wh} = 37.8 \text{ kWh}$$

$$\text{Cost} = 0.24 \times 37.8 = \$9.072$$

- 8 A PS4 costs \$399, inclusive of 10% GST. What was the original price of the PS4, before GST was added, correct to the nearest cent? 1

$$\$362.73$$

- 9 The stamp duty on a new vehicle is 3% of the first \$45 000 and 5% on the remainder of the marked price. Lulu buys a new vehicle with a marked price of \$55 000. What stamp duty must Lulu also pay? 1

$$\begin{aligned} \text{Stamp duty} &= (3\% \text{ of } \$45\,000) + (5\% \text{ of } \$10\,000) \\ &= (0.03 \times 45000) + (0.05 \times 10000) = \$1850 \end{aligned}$$

- 10 A washing machine has a power rating of 1500 watts. The machine is used in a self-service laundrette for 6 hours each day. The cost of running the machine is \$0.12 per kWh. What is the total cost of running the machine for 7 days? 1

$$\begin{aligned} \text{Number of hours used} &= 7 \times 6 = 42 \\ \text{Total energy used} &= 42 \times 1500 = 63000 \text{ Wh} = 63 \text{ kWh} \\ \therefore \text{Cost} &= 63 \times 0.12 = \$7.56 \end{aligned}$$

- 11 Allison uses a dishwasher with a rating of 800 W for 2.5 hours each day. Find the daily cost of electricity for using the dishwasher if electricity is charged at a rate of 21.40 cents per kWh. 2

$$\begin{aligned} 800 \text{ W} &= 0.8 \text{ kW} \\ \text{Energy used} &= 0.8 \times 2.5 = 2 \text{ kWh} \\ \text{Daily cost} &= 2 \times 0.214 \\ &= 0.428 \\ &= \$0.43 \text{ (nearest c)} \end{aligned}$$

- 12 The average flow rate of a normal shower head is 14 litres per minute. Harvey has one 5-minute shower each day. Calculate the annual cost of Harvey's showers given the water usage charge is \$2.05 per kilolitre. Answer to the nearest cent. 2

$$\begin{aligned} \text{Water used} &= 5 \times 14 \times 60 = 4200 \text{ L} = 4.2 \text{ kL} \\ \text{Annual cost} &= 4.2 \times 2.05 = 8.61 \\ \therefore \text{Annual cost} &\text{ is } \$8.61 \end{aligned}$$

- 13 Mary is 18 years old and has just purchased comprehensive motor vehicle insurance. The following excesses apply to claims for at-fault motor vehicle accidents. How much would Mary be required to pay as excess if she made a claim as the driver at fault in a car accident? 1

- Basic excess of \$850 for each claim
- An additional age excess of \$1600 for drivers under 25 years of age
- An additional age excess of \$400 for drivers 25 years of age or over with no more than 2 years driving experience

$$\text{Mary's excess} = 850 + 1600 \quad (\$400 \text{ does not apply as Mary is under 25.})$$

14 Reece is preparing his annual budget for 2006. His expected income is:

- \$90 every week as a swimming coach
- Interest earned from an investment of \$5000 at a rate of 4% per annum.

His planned expenses are:

- \$30 every week on transport
- \$12 every week on lunches
- \$48 every month on entertainment.

	A	B	C	D	E	F	G	H
1								
2								
3								
4								
5	Wages			\$4,680		Transport		\$ (Y)
6	Interest on investment			\$ (X)		Lunches		\$624
7						Entertainment		\$ (Z)
8								
9								

Reece will save his remaining income. He uses the spreadsheet below for his budget.

- a. Determine the values of X, Y and Z. (Assume there are exactly 52 weeks in a year.)

$$\text{Interest on Investment} = X \quad X = 5000 \times 4\% = \$200$$

$$\text{Transport} = Y \quad Y = 52 \times 30 = \$1560$$

$$\text{Entertainment} = Z \quad Z = 48 \times 12 = \$576$$

At the beginning of 2006, Reece starts saving.

- a. Will Reece have saved enough money during 2006 for a deposit of \$2100 on a car if he keeps to his budget? Justify your answer with suitable calculations.

$$\text{Total Income} = 4680 + 200 = \$4880$$

$$\text{Total Expenses} = 1560 + 624 + 576 = \$2760$$

$$\text{Savings} = 4880 - 2760 = \$2120$$

∴ Reece will have saved enough for a \$2100 deposit.

- 15 Margaret has a weekly income of \$900 and allocates her money according to the budget shown in the sector graph. How long will it take Margaret to save \$3600?



$$\text{Savings} = \frac{80}{360} \text{ sector graph} = \frac{80}{360} \times 900 = \$200 \text{ per week}$$

$$\therefore \text{Time to save } \$3600 = \frac{3600}{200} = 18 \text{ weeks}$$